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FOURTH SEMESTER B.Tech EXAMINATION, JULY – 2022

KINEMATICS AND THEORY OF MACHINES

Time:3 Hours **Maximum Marks:70**

Instructions:

i. Answer any FIVE full Questions. Questions 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1. a. Define inversion of a mechanism. Explain with the help of neat sketches the inversions of double slider crank chain. Give their applications. **07**

b. Explain Davis steering mechanism. **07**

2. a. Explain the Klein's construction for a single slider crank mechanism when a crank is rotating at uniform angular velocity. **04**

b. The kinematic diagram for one of the cylinders of a rotary engine is shown in Fig.2b **10**

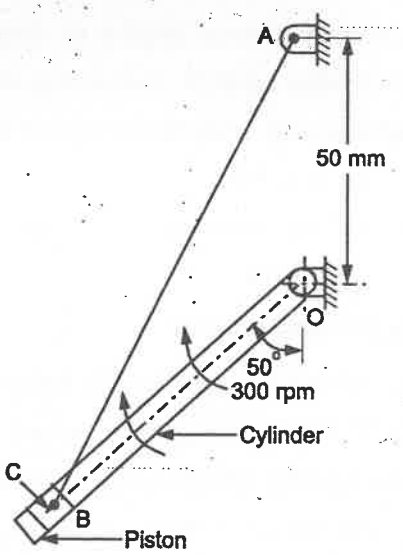

Fig. 2b Configuration diagram

The crank OA which is vertical and fixed, is 50 mm long. The length of the connecting rod is 125 mm. The line of the stroke OB is inclined at 50^0 to the vertical. The cylinders are rotating at a uniform speed of 300 rpm in a clockwise direction, about the fixed centre O. Determine:

i. Acceleration of the piston inside the cylinder.

ii. Angular acceleration of the connecting rod.

3. a. Explain with sketches, the different types of cams and followers **04**

b. A cam is to give the following motion to a knife-edged follower: **10**

i. Outstroke during 60° of cam rotation

ii. Dwell for the next 30° of cam rotation

iii. Return stroke during next 60° of cam rotation

Dwell for the remaining 210° of cam rotation.

The stroke of the follower is 40 mm and the minimum radius of the cam is 50 mm. The follower moves with uniform velocity during both the outstroke and return strokes. Draw the profile of the cam when (a) the axis of the follower passes through the axis of the cam shaft, (Graphical Method only)

OR

4. a. Explain turning moment diagram of single cylinder double acting steam engine. **05**

b. The turning moment diagram for a petrol engine is drawn to the following scales : Turning moment, 1 mm = 5 N-m ; crank angle, 1 mm = 1°. The turning moment diagram repeats itself at every half revolution of the engine and the areas above and below the mean turning moment line taken in order are 295, 685, 40, 340, 960, 270 mm^2. The rotating parts are equivalent to a mass of 36 kg at a radius of gyration of 150 mm. Determine the coefficient of fluctuation of speed when the engine runs at 1800 r.p.m. **09**

5. a. All the arms of a Porter governor are 178 mm long and are hinged at a distance of 38 mm from the axis of rotation. The mass of each ball is 1.15 kg and mass of the sleeve is 20 kg. The governor sleeve begins to rise at 280 r.p.m. when the links are at an angle of 30° to the vertical. Assuming the friction force to be constant, determine the minimum and maximum speed of rotation when the inclination of the arms to the vertical is 45°. **10**

b. What is the function of a governor? How does it differ from that of a flywheel? **04**

OR

6. a. What do you understand by gyroscopic couple? **07**

b. The turbine rotor of a ship has a mass of 3500 kg. It has a radius of gyration of 0.45 m and a speed of 3000 r.p.m. clockwise when looking from stern. Determine the gyroscopic couple and its effect upon the ship: **07**

1. when the ship is steering to the left on a curve of 100 m radius at a speed of 36 km/h.

2. when the ship is pitching in a simple harmonic motion, the bow falling with its maximum velocity. The period of pitching is 40 seconds and the total angular displacement between the two extreme positions of pitching is 12 degrees.

7. a. A, B, C and D are four masses carried by a rotating shaft at radii 100, 125, 200 and 150 mm respectively. The planes in which the masses revolve are spaced 600 mm apart and the mass of B, C and D are 10 kg, 5 kg, and 4 kg respectively. Find the required mass A and the relative angular settings of the four masses so that the shaft shall be in complete balance. **10**

b. Why is balancing of rotating parts necessary for high-speed engines? **04**

OR

8. a. The cranks and connecting rods of a 4-cylinder in-line engine running at 1800 r.p.m. are 60 mm and 240 mm each respectively and the cylinders are spaced 150 mm apart. If the cylinders are numbered 1 to 4 in sequence from one end, the cranks appear at intervals of 90° in an end view in the order 1-4-2-3. The reciprocating mass corresponding to each cylinder is 1.5 kg. **10**

Determine:

1. Unbalanced primary and secondary forces, if any

b. Write a short note on primary and secondary balancing. **04**